

Unitary Conjugacy Destroys Minimality of the Matrix Test-Function Families

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Claimed status: full negative resolution of both matrix-minimality questions in arXiv:1109.3795, subject to human review. For every $N \geq 2$, neither the normalized constrained-Hardy family $\tilde{\Psi}_1^N$ nor the normalized annulus family $\hat{\Psi}_{\mathbb{A}_q}^N$ is minimal. The scalar case $N = 1$ is known to be minimal, so both families are minimal exactly in the scalar case.

1 The two source questions

The paper asks whether the normalized matrix test family $\tilde{\Psi}_1^N$ for $\overline{\mathcal{B}}(H_1^\infty)^{N \times N}$ is minimal, and separately conjectures minimality of its normalized matrix annulus Agler decomposition. Here *minimal* has the sense used in the cited scalar work: no proper closed subfamily may be omitted while retaining the same class and decompositions.

unitary conjugacy minimal

$$\mathbf{U} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} : \begin{bmatrix} \mathcal{X} \\ \mathbb{C}^N \end{bmatrix} \rightarrow \begin{bmatrix} \mathcal{X} \\ \mathbb{C}^N \end{bmatrix}$$

so that S has the transfer-function realization

$$S(z) = D + C(I - \rho(\mathbb{E}(z))A)^{-1}\rho(\mathbb{E}(z))B.$$

Remark 4.6. An appealing conjecture is that the Agler decomposition (4.20) is *minimal* in the sense of [29] Section 5.1] and [30] Section 3.6].

4.2. The constrained Schur class over the unit disk. Following [26] [17], we define the *constrained Hardy space* H_1^∞ over the unit disk $\mathbb{D}$ to consist of bounded analytic functions s on $\mathbb{D}$ such that $s'(0) = 0$. One can check that this is still an

Figure 1: The annulus minimality conjecture on PDF page 32.

2 A general symmetry obstruction

For a matrix-valued test $\psi : \Omega \rightarrow M_N$ and a constant unitary $U \in \mathcal{U}(N)$, write

$$\psi^U(z) = U\psi(z)U^*.$$

Lemma 1 (Conjugate tests give the same constraint). *For every positive operator-valued kernel K , the test ψ is a contractive right multiplier on the source’s Hilbert spaces if and only if ψ^U is. Consequently, ψ and ψ^U impose identical admissibility constraints.*

Idea of the proof

The paper tests contractivity against *every* Hilbert–Schmidt-valued function X . Conjugating ψ merely changes that dummy variable from X to U^*X .

of 0 with respect to this triangle, or (2) a pair of antipodal points on the unit circle with weights then necessarily $(\frac{1}{2}, \frac{1}{2})$. When the reduction described in the previous paragraph is carried out, one restricts to triples of points $\mathbf{t} = (1, t_2, t_3)$ which include 1 and there is only one antipodal pair of points $(1, -1)$. These authors also show that this space Θ with its natural topology is homeomorphic to the unit sphere. They also show that the collection $\tilde{\Psi}_1^1$ is a minimal collection of test functions for $\overline{\mathcal{B}}H_1^\infty$. Whether $\tilde{\Psi}_1^N$ is a minimal collection of test functions for $\overline{\mathcal{B}}(H_1^\infty)^{N \times N}$ in general we leave as an open question.

As we have seen, there is a dual issue of finding minimal generating sets for admissible collections of kernels $\mathcal{K}_\Psi(\mathbb{C}^N)$, as well as finding small generating sets for

Figure 2: The constrained-Hardy open question on PDF page 36.

Proof. Theorem 2.4 of the source says that admissibility is equivalent to positivity, for every X , of

$$k_{X,\psi,K}(z, w) = \text{tr}[X(w)^*(I - \psi(w)^*\psi(z))X(z)K(z, w)]. \quad (1)$$

Since

$$I - \psi^U(w)^*\psi^U(z) = U(I - \psi(w)^*\psi(z))U^*,$$

we have

$$k_{X,\psi^U,K} = k_{U^*X,\psi,K}.$$

The substitution $X \leftrightarrow U^*X$ is a bijection on the class of all test functions X , proving the equivalence. $\square$

Theorem 2 (Symmetry obstruction). *Let Ψ be a Hausdorff topological family of M_N -valued test functions. Assume that $\psi \mapsto \psi^U$ is a homeomorphism of Ψ and that $\psi_0^U \neq \psi_0$ for some $\psi_0 \in \Psi$. Then Ψ is not minimal: there is a proper closed $C \subsetneq \Psi$ such that*

$$\mathcal{K}_C(\mathcal{E}) = \mathcal{K}_\Psi(\mathcal{E})$$

for every auxiliary Hilbert space $\mathcal{E}$. Hence $\mathcal{SA}_C = \mathcal{SA}_\Psi$ and every general Agler-decomposition theorem valid for Ψ is valid with C .

Proof intuition

Take a small open neighborhood of ψ_0 disjoint from its conjugate and delete it. Every deleted test still has a conjugate outside the deleted set, and Lemma 1 says that this retained conjugate imposes the very same constraint.

Proof. Choose disjoint open neighborhoods V of ψ_0 and W of ψ_0^U , and put

$$O = V \cap \{\psi : \psi^U \in W\}, \quad C = \Psi \setminus O.$$

Then O is nonempty and open, so C is proper and closed. Moreover $\psi^U \in W \subset C$ for every $\psi \in O$.

Certainly $\mathcal{K}_\Psi(\mathcal{E}) \subseteq \mathcal{K}_C(\mathcal{E})$. Conversely, if $K \in \mathcal{K}_C(\mathcal{E})$ and $\psi \in \Psi$, either $\psi \in C$, or $\psi \in O$ and the equivalent test ψ^U lies in C . In both cases K satisfies the ψ constraint by Lemma 1. Thus $K \in \mathcal{K}_\Psi(\mathcal{E})$, proving equality. The Schur–Agler class is defined solely through these admissible kernels, so it is unchanged. $\square$

3 The constrained-Hardy family

Lemma 3. *For every $N \geq 2$, $\tilde{\Psi}_1^N$ is invariant under constant unitary conjugation and contains a noncentral test.*

Proof. Unitary conjugation preserves the normalized constrained matrix Herglotz class, its extreme points, the Cayley transform, and the normalization $S(1) = I_N$. Hence it acts homeomorphically on $\tilde{\Psi}_1^N$.

We give an explicit noncentral member. Let P_j be the coordinate rank-one projections. For the first coordinate use the scalar measure

$$\mu_1 = \frac{1}{2}(\delta_1 + \delta_{-1});$$

for the second use

$$\mu_2 = \frac{1}{3}(\delta_{e^{i\pi/6}} + \delta_{e^{i5\pi/6}} + \delta_{e^{i3\pi/2}}).$$

Both are probability measures with first moment zero, and both are extreme: on their two or three support points, the mass and complex first-moment equations uniquely determine the weights. Their supports are disjoint. Thus

$$\mu = \mu_1 P_1 + \mu_2 P_2$$

is an extreme 2×2 constrained quantum probability measure. Indeed, in any positive convex decomposition of μ , vanishing off its support forces both summands onto that support; at each atom, rank-one support forces the summand onto the same coordinate range; the two scalar extremality statements then force equality.

The scalar Herglotz functions and Cayley transforms are

$$\frac{1+z^2}{1-z^2} \longleftrightarrow z^2, \quad \frac{i+z^3}{i-z^3} \longleftrightarrow -iz^3.$$

After the source's right-unitary normalization at 1, the resulting member of $\tilde{\Psi}_1^2$ is

$$\psi_0(z) = \text{diag}(z^2, z^3).$$

For $N > 2$, append scalar extreme measures on pairwise-disjoint rotated three-point supports. This gives a normalized test whose first two entries are still z^2, z^3 . Swapping the first two coordinates gives a distinct unitary conjugate, so the test is noncentral. $\square$

Corollary 4 (Constrained-Hardy question). *For every $N \geq 2$, $\tilde{\Psi}_1^N$ is not a minimal collection of test functions for $\overline{\mathcal{B}}(H_1^\infty)^{N \times N}$.*

Proof. Apply Theorem 2 and Lemma 3. $\square$

4 The annulus family

The reduced annulus tests in the source have the explicit form

$$\Phi_{[V],t}(z) = V \text{diag}(\varphi_{t_1}(z), \dots, \varphi_{t_N}(z))V^*,$$

and satisfy $\Phi_{[V],t}(1) = I_N$. The family $\widehat{\Psi}_{\mathbb{A}_q}^N$ is therefore invariant under conjugation: conjugation by U replaces V by UV . For $N \geq 2$, choose t_1, t_2 for which the scalar annulus tests are distinct. Taking $V = I$ produces a noncentral member.

Corollary 5 (Annulus conjecture). *For every $N \geq 2$, the matrix annulus family $\widehat{\Psi}_{\mathbb{A}_q}^N$ is not minimal. In particular, the appealing conjecture in Remark 4.6 is false in every genuinely matrix-valued size.*

Proof. Apply Theorem 2. The resulting proper closed subfamily has the same admissible kernels, hence the source's general theorem supplies the same Agler decompositions using that subfamily. $\square$

5 Conclusion and scope

The source cites scalar minimality for both examples. Corollaries 4 and 5 therefore give the exact classification:

each normalized family is minimal if and only if $N = 1$.

The obstruction concerns the source's families as stated. It suggests the natural repair: first quotient constant-unitary conjugacy, or choose a topological slice for that action, before asking a refined minimality question.

A bounded search of the run registry, exact question wording, title, citations, and unitary-conjugacy terminology found no later explicit resolution. Novelty is provisional; the proof is self-contained.

References

- [1] J. A. Ball and M. D. Guerra Huamán, *Test functions, Schur–Agler classes and transfer-function realizations: the matrix-valued setting*, arXiv:1109.3795; Complex Analysis and Operator Theory **7** (2013), 529–575, doi:10.1007/s11785-012-0218-x.
- [2] M. A. Dritschel and J. Pickering, *Test functions in constrained interpolation*, Transactions of the American Mathematical Society **364** (2012), 5589–5604.